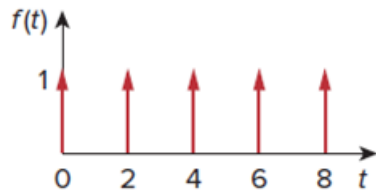


Problem

Calculate the Laplace transform of the infinite train of unit impulses in Fig. 15.31.

Figure 15.31:



Step-by-step solution

Step 1 of 4

The following is the given train of unit impulses:

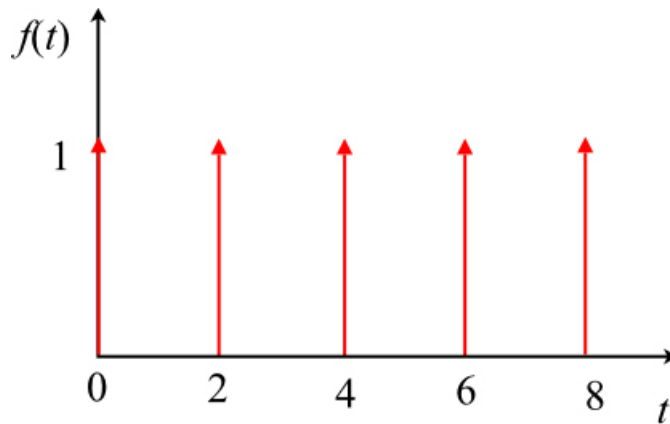


Figure 1

Step 2 of 4

From Figure 1, it is known that the function $f(t)$ is a periodic function with period $T = 2$.

Hence,

$$\begin{aligned} f(t) &= f(t + nT) \\ &= f(t + n(2)) \quad (\text{Since, } T = 2) \\ &= f(t + 2n) \end{aligned}$$

Step 3 of 4

From Figure 1,

$$f_1(t) = \delta(t)$$

Apply Laplace transform.

$$\begin{aligned} F_1(s) &= L\{\delta(t)\} \\ &= 1 \end{aligned}$$

Step 4 of 4

It is known that,

Laplace transform of a periodic function $f(t) = f(t + nT)$ is $F(s) = \frac{F_1(s)}{1 - e^{-Ts}}$. Hence the

Laplace transform of given function $f(t) = f(t + 2n)$ is

$$\begin{aligned} F(s) &= \frac{F_1(s)}{1 - e^{-2s}} \\ &= \frac{1}{1 - e^{-2s}} \quad (\text{Since, } F_1(s) = 1) \end{aligned}$$

Thus the Laplace transform of the given train of unit impulses is $\boxed{\frac{1}{1 - e^{-2s}}}$.